IEAP-Rstudio-Series03: Data wrangling and ANOVA

LE ROUX Jeanne, WAHEED Ayesha

2026-10-12

Repository: https://github.com/ayeshawaheed9/IEAP-Series03-Rstudio

1 ANOVA: Analysis of variance

We want to reproduce Fig 3 of the article: the effect of difficulty (ID) on movement time (MT) in three groups (ap, cp, pp).

Section: 1.1

1. **Is the effect of difficulty the same in all groups?** We look at the **GROUP x ID interaction**. If it is significant, the effect of difficulty is different between groups (for example, much bigger in one group than in the others). If it is not significant, difficulty has about the same effect in all groups.
2. **How do we visualize this?** We plot movement time (MT, y axis) as a function of difficulty (ID, x axis), with one line per group. If the lines are parallel, the effect is the same in all groups. If the slopes are different, the effect is different. The slope of each regression line tells us how many seconds MT increases when ID increases by one bit.
3. **Is there a significant difference?** We read the p-values of the ANOVA. If $p < .05$, the difference is unlikely to be due to chance. We look at three effects: GROUP (do the groups differ overall?), ID (does difficulty change MT?) and GROUP x ID (does difficulty change MT differently in each group?).
4. **What sentence do we write in the report?** We write one short sentence for each result of the ANOVA, with its numbers in brackets. The numbers are F, the degrees of freedom and the p-value. For example: "Movement time increased with difficulty ($F = \ldots$, $p < \ldots$). This increase was different between the groups ($F = \ldots$, $p < \ldots$)." The final sentence with our real numbers is in section 1.3.

GROUP is a between-subject factor and ID is a within-subject factor, so we need a **mixed ANOVA**.

1.2 What library to answer the above questions?

- `tidyverse` to reorganize the data and plot
- `ez` to run the ANOVA

```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.3.3
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.5
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   3.5.2     v tibble    3.2.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(ez)
```

```
Warning in check_dep_version(): ABI version mismatch:
lme4 was built with Matrix ABI version 1
Current Matrix ABI version is 0
Please re-install lme4 from source or restore original 'Matrix' package
```

1.3 Data reorganization

```
df <- read_table("data/Results.txt", show_col_types = FALSE) %>%
  rename(MT = MovementTime) %>%
  mutate(GROUP = factor(GROUP), SUBJ = factor(SUBJ),
         ID = factor(ID), REP = factor(REP)) %>%
  as.data.frame()
```

Count the data points for each combination of factors, to look for missing data.

```
df %>% count(GROUP, ID)
```

```
  GROUP  ID    n
1    ap ID1  740
2    ap ID2  740
3    ap ID3  740
4    cp ID1 1140
5    cp ID2 1140
6    cp ID3 1140
7    pp ID1  760
8    pp ID2  760
9    pp ID3  400
```

Data for groups is imbalanced for ID3 in the group pp. We remove ID3.

```
df <- df %>% filter(ID != "ID3") %>% droplevels()
levels(df$ID)
```

```
[1] "ID1" "ID2"
```

```
df %>% count(GROUP, ID)
```

```
  GROUP  ID    n
1    ap ID1  740
2    ap ID2  740
3    cp ID1 1140
4    cp ID2 1140
5    pp ID1  760
6    pp ID2  760
```

The data are still unbalanced. Let's check REP.

```
df %>% count(REP)
```

```
  REP    n
1  R1 2040
2  R2 2040
3  R3 1200
```

```r
df %>% count(GROUP, REP)
```

```
  GROUP REP   n
1    ap  R1 520
2    ap  R2 520
3    ap  R3 440
4    cp  R1 760
5    cp  R2 760
6    cp  R3 760
7    pp  R1 760
8    pp  R2 760
```

Data imbalanced for REP3 in the groups pp and ap. We remove REP3.

```r
df <- df %>% filter(REP != "R3") %>% droplevels()
df %>% count(GROUP, REP)
```

```
  GROUP REP   n
1    ap  R1 520
2    ap  R2 520
3    cp  R1 760
4    cp  R2 760
5    pp  R1 760
6    pp  R2 760
```

Number of participants per group:

```r
df %>% distinct(GROUP, SUBJ) %>% count(GROUP, name = "count") %>% as.data.frame()
```

```
  GROUP count
1    ap    13
2    cp    19
3    pp    19
```

One value per participant and ID

The ANOVA wants a single value per participant and per condition: we make the average.

```r
data_anova <- df %>%
  group_by(GROUP, SUBJ, ID) %>%
  summarise(MT = mean(MT), .groups = "drop")
```

ANOVA

```r
anova_res <- ezANOVA(data = data_anova,
                     dv = MT,
                     wid = SUBJ,
                     within = ID,
                     between = GROUP,
                     detailed = FALSE)
```

```
Warning: Data is unbalanced (unequal N per group). Make sure you specified a
well-considered value for the type argument to ezANOVA().
```

```r
anova_res$ANOVA
```

```
    Effect DFn DFd        F            p p<.05        ges
2    GROUP   2  48 37.24408 1.720007e-10     * 0.57462110
3       ID   1  48 30.77618 1.221240e-06     * 0.07667596
4 GROUP:ID   2  48 11.58124 7.866206e-05     * 0.05882293
```

Movement time increased with difficulty, $F(1, 48) = 30.78$, $p < .001$. This increase was different between the groups, $F(2, 48) = 11.58$, $p < .001$.

1.4 Run the linear regression by GROUP

```r
data_reg <- data_anova %>%
  mutate(ID_num = parse_number(as.character(ID)))

reg_table <- data_reg %>%
  group_by(GROUP) %>%
  group_modify(~ {
    m <- lm(MT ~ ID_num, data = .x)
    tibble(intercept = coef(m)[1], slope = coef(m)[2])
  }) %>%
```

```r
  mutate(label = sprintf("MT = %.3f + %.3f * ID", intercept, slope))

reg_table
```

```
# A tibble: 3 x 4
# Groups:   GROUP [3]
  GROUP intercept slope label
  <fct>     <dbl> <dbl> <chr>
1 ap        0.730 0.269 MT = 0.730 + 0.269 * ID
2 cp        0.634 0.201 MT = 0.634 + 0.201 * ID
3 pp        1.69  1.46  MT = 1.689 + 1.464 * ID
```

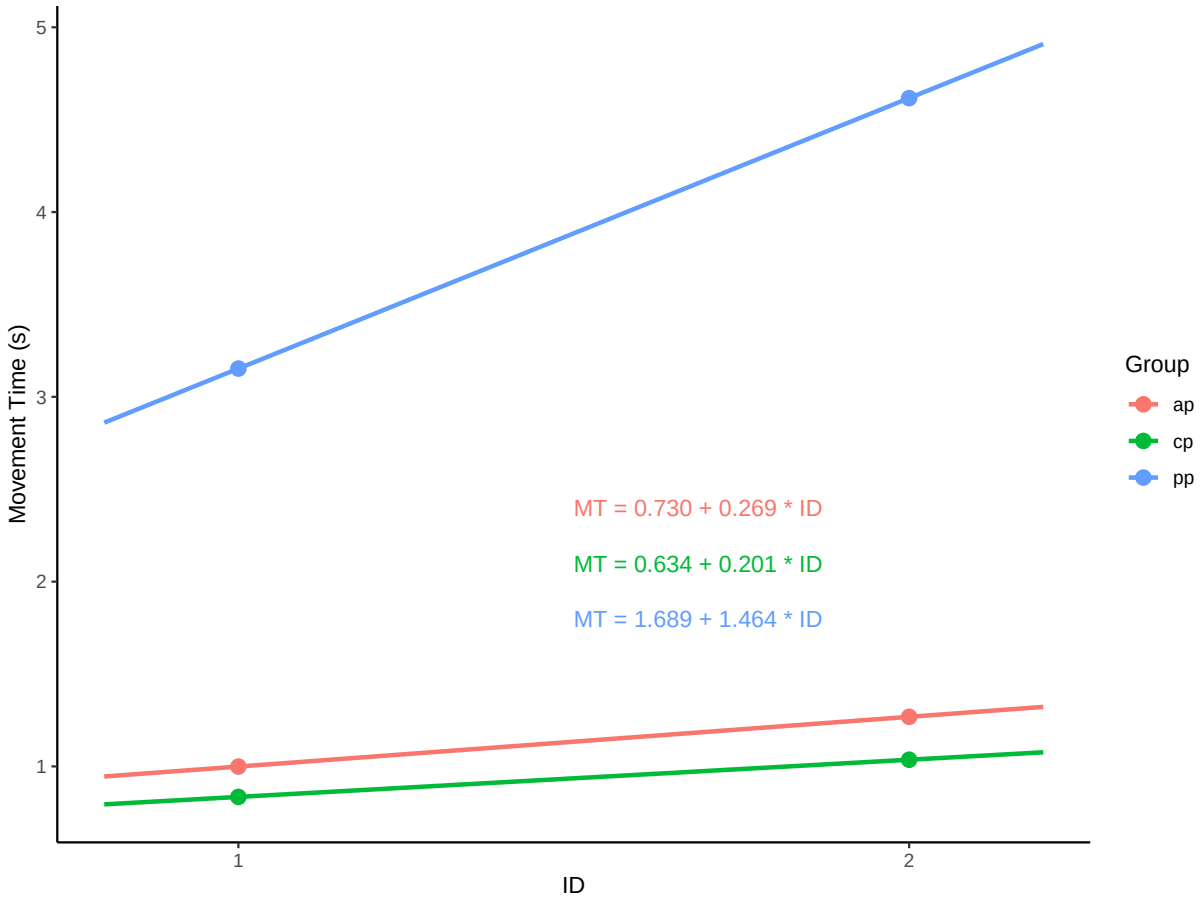
1.5 Graph for publication

```r
means <- data_reg %>%
  group_by(GROUP, ID_num) %>%
  summarise(MT = mean(MT), .groups = "drop")

eq_pos <- reg_table %>%
  ungroup() %>%
  mutate(x = 1.5, y = 2.7 - 0.3 * row_number())

p <- ggplot(means, aes(x = ID_num, y = MT, colour = GROUP)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE,
              fullrange = TRUE) +
  geom_text(data = eq_pos, aes(x = x, y = y, label = label),
            hjust = 0, show.legend = FALSE) +
  scale_x_continuous(breaks = c(1, 2), limits = c(0.8, 2.2)) +
  labs(x = "ID", y = "Movement Time (s)", colour = "Group") +
  theme_classic()

p
```



1.6 Export the graph as a PDF

```
dir.create("figures", showWarnings = FALSE)
ggsave("figures/fig3_MT_ID.pdf", plot = p, width = 8, height = 6,
       units = "in")
```

1.7 Discuss the differences with the article

Our regression equations (e.g., MT = 0.730 + 0.269 × ID for ap) differ from those in the article. Our method is simpler: we averaged the data per participant and per ID, then ran a classic ANOVA. Nevertheless, the GROUP × ID interaction is significant, $F(2, 48) = 11.58$, $p < .001$, so the conclusion is the same: the effect of difficulty depends on the group. The simple method keeps the main pattern of the data: a much steeper increase of movement time with ID in the pp group than in the ap and cp groups. The more sophisticated method of the article changes the exact values (slopes and intercepts), but not this pattern.

Git workflow analysis

How we organised the work

The repository was created by Ayesha and is public. We split the assignment in two big parts, each on its own branch:

- `Data_loading_and_cleaning` (Ayesha): sections 1.1 to 1.3, data loading and cleaning.
- `Anova_and_data` (Jeanne): sections 1.3 to 1.7, mean per participant, ANOVA, regression by group, figure, PDF export and discussion.

The other parts of the report were then written one by one, each on its own branch: `readme`, `sources`, `git-workflow`, `challenges`, `checklist`, `final-report`and `checklist-headings`. Jeanne wrote these parts, since Ayesha, who owns the repository, reviewed and merged them. At the end of the project, the repository contains one branch per part, each merged through a Pull Request reviewed by the other member.

The steps we followed for each branch

1. Update `main`, then create the branch from it.
2. Work in a sub-document (for example `02-anova_and_data.qmd`) and make small commits, one per question or per chunk, with a message that says what the commit does.
3. Push the branch and open a Pull Request with a short description.
4. The other person reviews the Pull Request and leaves comments.
5. We implement the requested changes, then Ayesha merges the Pull Request into `main`.

The master document includes the sub-documents with `{{< include file.qmd >}}`, so each part could be written without touching the work of the other person.

What the reviews changed

- During the first Pull Request, Jeanne pointed out that the data for ID3 in group pp were not completely missing (400 observations were available). Ayesha implemented the change before the merge.
- During the second Pull Request, Ayesha noticed that Jeanne's commit messages contained the Git commands and not only the message. The 8 commit messages were rewritten before the merge, so the history of `main` is readable.

What worked and what we would do differently

What worked: working on separate branches prevented any conflict, and the Pull Requests forced us to explain our changes and to check each other's work.

What we would do differently: create smaller branches from the start (one per question and not one per big part), so that the history shows at least one commit per question for both of us, and agree on the names of the R objects (`df`, `data_anova`) before writing the code.

Challenges and lessons learned

Challenges with the data and the statistics

- **Unbalanced data.** ID3 was incomplete in group pp, and R3 was missing in groups ap and pp. We removed both so that all groups had the same conditions. *Lesson:* always count the observations by condition before running an ANOVA.
- **Different names between the two parts.** The cleaned data were called `df` with the column `MT`, while the code of the analysis had been written with other names. *Lesson:* agree on object and column names before splitting the work.
- **Different F value for ID.** We obtained $F(1, 48) = 30.78$ for the effect of ID, while the assignment gives 26.11. The groups have different numbers of participants (13, 19 and 19), which changes how this effect is computed. The conclusion does not change.
- **Regression equations.** Our equations (for example $MT = 0.730 + 0.269 \times ID$ for ap) do not match those of the assignment. We tested a regression on the mean per participant and on all the trials, and neither gave the same equations. *Lesson:* a simpler method can give the same conclusion with different numbers, which is what we explain in section 1.7.
- **Equations written on top of each other on the graph.** The three equations had the same height because the table was still grouped. Adding `ungroup()` fixed it.

Challenges with Git

- **Commands written in the commit message.** We pasted `git add ... && git commit -m "..."` in the message field. We rewrote the messages with an interactive rebase and sent them with `git push --force-with-lease`. *Lesson:* the commands go in the terminal, only the message goes in the commit; and a history can only be rewritten before the merge, never on `main`.
- **Terminal stuck in the editor.** The rebase opened the editor `nano`, and our commands were typed inside it. We learned to leave it with `Control + X`, `Y` and `Enter` (the `Control` key, not `Cmd`).

- **Nothing to commit.** We committed a file that was not saved yet, so Git answered "nothing to commit". *Lesson:* check that the star next to the file name has disappeared before committing.
- **File saved in the wrong folder.** A new file was saved in another project, so Git did not see it. We moved it into the repository. *Lesson:* check the folder in the save window.
- **Working from the right branch.** We learned to update `main` with a pull before creating a new branch, and to check the name of the current branch in the Git panel.
- **Wrong branch name.** We created a branch called `Git_workflow` and tried to push `git-workflow`, so Git answered "src refspec does not match any". *Lesson:* Git is case-sensitive; check the name with `git branch` and rename the branch with `git branch -m`.

What we learned

- A clear Git history needs small commits with meaningful messages and at least one commit per question or per chunk of code.
- Branches and Pull Requests let two people work in parallel and review each other's work.
- Reading an error message calmly, and asking what the tool is doing, solved most of our problems.

Data sources

- **`data/Results.txt`**: movement time data used to reproduce Fig 3 of the article below (groups ap, cp and pp; difficulty levels ID1 to ID3; three repetitions). Source: Moodle course "M1: Python, R and Git for data analysis", Series 03. URL: https://moodle.umontpellier.fr/mod/folder/view.php?id=1025707

References

1. Mottet D, van Dokkum LEH, Froger J, Gouaïch A, Laffont I. Trajectory formation principles are the same after mild or moderate stroke. *PLOS ONE*. 2017 (published 22 March 2017). DOI: 10.1371/journal.pone.0173674. Fig 3: 10.1371/journal.pone.0173674.g003.
2. Wickham H, Averick M, Bryan J, et al. Welcome to the tidyverse. *Journal of Open Source Software*. 2019;4(43):1686. DOI: 10.21105/joss.01686.
3. Lawrence MA. *ez: Easy Analysis and Visualization of Factorial Experiments* (R package). Available at https://CRAN.R-project.org/package=ez (no DOI).

Checklist

This checklist follows the assignment and the grading criteria.

Repository

- ☒ Public GitHub repository, with the `main` branch up to date
- ☒ Repository named `IEAP-Series03-Rstudio`, as in the grading criteria
- ☒ Clear and complete `README.md`
- ☒ `LICENSE` file (GPL-3.0)
- ☒ Master document named `IEAP-Series03-Rstudio.qmd`
- ☒ Corresponding PDF file in the repository

Git history and content of the report

- ☒ Meaningful commit messages (only the message, without the Git commands)
- ☒ Regular commits with meaningful messages, at least one commit per part of the report
- ☒ One branch per part of the work, merged through a Pull Request reviewed by the other member
- ☒ Authors and date in the header
- ☒ Clickable link to the GitHub repository
- ☒ Analysis of our workflow strategy using Git
- ☒ Analysis of the challenges encountered and lessons learned
- ☒ Complete code and answers to the questions (1.1 to 1.7)
- ☒ Data sources, with URLs
- ☒ Scientific references, with DOIs
- ☒ This checklist
- ☒ The master document includes all the sub-documents when rendering
- ☒ PDF file uploaded on Moodle